

Failure Modes and Fault Signatures

Stamping press 01 (press-stamp-01) | Main drive and lubrication circuit

Machine	Stamping press 01 (machine ID press-stamp-01, type press, area Press shop, site vit-lab), stamping press in a car plant
Sensors	MAIN_MOTOR_CURRENT (A), LUBE_OIL_PRESSURE (bar), BEARING_VIBRATION_RMS (mm/s) and the 64-bin spectrum BEARING_SPECTRUM (mm/s); sensor IDs PRESS-STAMP-01.<CHANNEL_KEY>
Components / subsystem	Press main drive and its pressure lubrication circuit: main drive motor, lubrication unit (gear pump, reservoir, lube filter) and main bearing housing; planned agent tags main_drive_motor, lube_filter, lubrication_unit, main_bearing_housing
Document title	Failure Modes and Fault Signatures
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Purpose	Explains how faults of the press main drive and lubrication circuit appear in the telemetry of press-stamp-01 and how the telemetry separates them: pattern, evidence, severity, inspection, action and post-repair behaviour.
Scope	Two simulated faults of the repository (CLOGGED_FILTER, BEARING_WEAR) in full numerical detail, plus engineering-reference failure modes that are not modelled. Limits and levels are in Document 01, the procedure in Document 03, maintenance in Document 04 and cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_PRESS_STAMP_01.md (catalog limits, simulation equations, topics, runbook stubs, tests).
[D]	Derived value: computed from the repository's own simulation model (seed 1, 2 Hz, 600 s per case) or algebraically from its equations, as recorded in the context file.
[E]	Engineering reference: general industrial practice for press drives, lubrication circuits and vibration diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. There is no physical press or transducer. An ESP32 gateway (or a software simulator) plays the press controller and generates values from documented equations. The repository describes its limits as illustrative constants chosen for the demo, not vendor values, and its runbooks as not an OEM manual. Values tagged [P] are therefore prototype values, not OEM specifications; values tagged [E] and [I] are not manufacturer-certified.

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1. Purpose and Method

1.1 What this document contains

The repository models exactly two faults for press-stamp-01: CLOGGED_FILTER and BEARING_WEAR [P]. They share a symptom, high bearing vibration, and differ in evidence. This document explains how to tell them apart from the telemetry and adds engineering-reference failure modes (F03 to F16) so that the diagnosis can say "a different fault" when the evidence does not fit. The repository states that no other press failure mode and no combination of the two faults is modelled [P]; every profile for F03 to F16 is therefore [E] and its numbers, where given, are [I].

RULE: a fault name from the repository (CLOGGED_FILTER, BEARING_WEAR) is a simulation scenario. The scenario label is hidden ground truth (diag.sim) and must not be used as evidence; reason from channel evidence [P].

1.2 Evidence vector used in every profile

Table 1. Evidence vector

Evidence	Symbol	Healthy	Source of the value
Lube oil pressure, 60 s mean	P60	4.50 bar	LUBE_OIL_PRESSURE [P]
Vibration RMS, 60 s mean	V60	1.31 mm/s	BEARING_VIBRATION_RMS [P]
Motor current, 60 s mean and stroke peak	I60, Ipk	94.8 A, about 150 A	MAIN_MOTOR_CURRENT [P] / [D]
Mean bin amplitude above 300 Hz	HB300	0.030 mm/s	BEARING_SPECTRUM [P]
Peak-to-median above 300 Hz	PMR300	not specified; below 2.5 flat, above 3 peaked	tests [P]
Bearing defect lines (bins 7, 14, 21, 28, 35)	DL	Healthy amplitude	tests [P]
1X and 2X amplitudes	A1X, A2X	1.05 and 0.45 mm/s	spectrum [P]
Data quality	DQ	status online, no gaps	MQTT contract [P]

2. Failure-Mode Classification

Table 2. Fault library

ID	Fault	Component	Class	In simulator	Primary evidence	Severity class
F01	CLOGGED_FILTER (clogged lube filter)	Lube filter, lubrication unit	Lubrication	Yes [P]	P60 falls, V60 rises, broadband floor above 300 Hz	Medium, high if ignored
F02	BEARING_WEAR (main bearing outer-race defect)	Main bearing housing	Mechanical	Yes [P]	V60 rises slowly, discrete harmonic family, P normal	High (days of downtime)
F03	Lube pump wear or failure	Gear pump	Lubrication	No [E]	P falls, pulsation changes, filter fault excluded	High
F04	Oil leak or low reservoir level	Reservoir, lines	Lubrication	No [E]	P falls irregularly, aeration noise	High
F05	Blocked nozzle or line after the transducer	Distribution line, nozzle	Lubrication	No [P] named, not modelled	P rises above normal	Medium
F06	Relief valve stuck or setpoint drift	Relief valve	Lubrication	No [E]	P steady low or high step	Medium
F07	Oil temperature or viscosity effect	Oil	Operational	No [E]	P follows temperature, slow cycle	Low
F08	Main drive motor or supply fault	Main drive motor	Electrical	No [E]	I rises or ripples, line-frequency lines	High
F09	Process or die overload	Press, die	Operational	No [E]	Ipk rises, V swells in working stroke	Medium
F10	Imbalance or misalignment	Shaft, coupling	Mechanical	No [E]	1X and 2X rise, integer orders	Medium
F11	Mechanical looseness	Bearing housing, foundation	Mechanical	No [E]	Many harmonics of 1X, sub-harmonics	Medium to high
F12	Pressure transducer fault	LUBE_OIL_PRESSURE sensor	Sensor	No [E]	P flat, offset or drifting without V change	Low (data)
F13	Vibration sensor or mounting fault	BEARING_VIBRATION_RMS sensor	Sensor	No [E]	V spikes, flat or implausible; RMS and spectrum disagree	Low (data)
F14	Current sensor fault	MAIN_MOTOR_CURRENT sensor	Sensor	No [E]	I flat or offset, pulse missing	Low (data)
F15	Data-path or gateway fault	ESP32 gateway, broker, backend	Data	Partly [P]	stale, offline, seq gaps, validation drops	Low (data)
F16	Combined clog and bearing wear	Both	Combined	No [P]	Falling P plus discrete lines on a broadband floor	High

2.1 Reading the classification

- F01 and F02 carry [P] numerical signatures; use them as the reference patterns.
- F03 to F06 are lubrication faults that move LUBE_OIL_PRESSURE; F01 differs by being the only one whose pressure fall is accompanied by a flat broadband floor in the spectrum [E].
- F08 to F11 move current or the spectrum without a pressure fall.
- F12 to F15 are measurement or data faults and are always checked first (Document 03, step 2).
- F16 is not modelled: the agent must treat mixed evidence as lubrication first [P].

3. Signature Consistency Ratios

Because the simulated faults follow fixed equations, the telemetry of each fault obeys relations that a measurement fault, a different physical fault or a mix of faults does not. The relations below are [D]; they are tested as consistency checks in Document 03.

Table 3. Equations and the relations they imply [P] / [D]

Quantity	CLOGGED_FILTER	BEARING_WEAR
Pressure	$P = 4.5 \times (1 - 0.45\ r) + 0.06\ r \times \text{noise}$; falls to 2.47 bar	Unchanged
Current	$I \times (1 + 0.03\ r)$: mean 94.8 to 97.6 A	$I \times (1 + 0.08\ r)$: mean 94.8 to 102.3 A (max 164.1 A)
Vibration	Floor above 300 Hz: RMS scales by $(1 + 2.75\ r)$; V60 about $1.31 + 3.68\ r$ (4.98 at $r = 1$)	Lines at 3.55 to 17.75 x 1X with sidebands: V60 about $1.31 + 1.58\ r$ (2.89 at $r = 1$)
Spectrum shape	Flat floor, 18 percent jitter per bin, no discrete lines	Five lines with weights 1.0, 0.8, 0.6, 0.45, 0.3 and sidebands at 0.4 of the carrier
Delta V60 / Delta P60	-1.8 mm/s per bar	Undefined
Delta I60 / Delta V60	0.77 A per mm/s	4.8 A per mm/s
HB300	0.030 to 0.749 mm/s	Small change; lines at 351 and 439 Hz
Alarm reached at full severity	Yes: V above 4.0 mm/s and P below 3.0 bar	No: V peaks about 3.2 mm/s; P normal

Table 4. Consistency checks for the diagnosis [I]

Check	Passes for	Fails means
Estimated r from P60 within 0.15 of r from V60 (clog formulae)	CLOGGED_FILTER	Not a pure clog: pressure sensor fault, other lubrication fault or mixed
Estimated r from V60 within 0.15 of r from I60 (bearing formulae)	BEARING_WEAR	Not a pure bearing fault: process or drive fault, current sensor fault
RMS equals the square root of the summed squared spectrum bins	Any valid data	Decoding or data fault (F13, F15)
V60 rise matches HB300 rise (broadband) for a clog	CLOGGED_FILTER	Vibration sensor fault or another cause

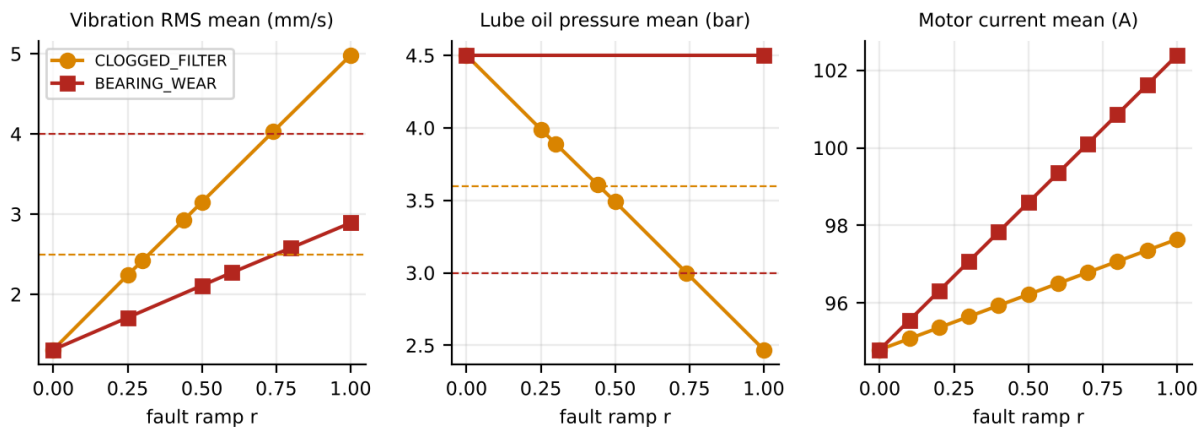


Figure 1. Progression of vibration, pressure and current means against the fault ramp r for the two simulated faults (derived table [D]); dashed lines are the warn and alarm limits [P].

4. Detailed Fault Profiles

4.1 F01 CLOGGED_FILTER: clogged lubrication filter

Table 5. F01 profile

Attribute	Description
Fault	CLOGGED_FILTER (clogged lube filter)
Affected component	Lube filter in the lubrication unit; consequence at the main bearing housing
Physical mechanism	Filter media loads with debris; the pressure transducer sits downstream of the filter, so the reading falls. A starved bearing loses its oil film and vibration rises [P]
Sensor behaviour	P falls up to 45 percent (4.5 to about 2.47 bar). V rises, 4.98 mm/s mean at $r = 1$. Current up about 3 percent. Spectrum gains a broadband floor above 300 Hz [P]
Expected telemetry pattern	Smooth monotonic fall of P60 with a simultaneous rise of V60 and HB300; no discrete lines; I60 rises about 2.8 A per unit r [D]
Thresholds crossed	V samples reach warn from r about 0.25; V60 reaches 2.5 at r about 0.33; P60 reaches 3.6 bar at r about 0.44; P60 3.0 bar and V60 4.0 mm/s at r about 0.74 [P] / [D]
Severity	Medium when found early (a short cartridge swap); High if ignored (bearing damage, unplanned stop of days) [P]
Likely causes	Dirty oil, filter at end of life, wrong or poor cartridge, contamination ingress; develops over minutes to days depending on oil cleanliness [P]
Symptoms	Rising bearing noise, pressure gauge reading low (if fitted), warning on vibration and pressure [E]
Alternative explanations	Pump wear (F03), leak (F04), relief valve (F06), transducer fault (F12), cold-oil or hot-oil effect (F07) [E]
Differentiating evidence	Pressure falls while the spectrum floor is flat and broadband above 300 Hz (PMR300 below 2.5) and current rises only about 3 percent [P]
Inspection requirements	Confirm P trend over the last hour, spectrum shape and that other channels are normal; check filter cartridge [P]
Corrective action	Replace the filter cartridge (a short task, in the order of minutes); follow the OEM procedure with lock-out and tag-out and relieve line pressure [P]
Expected post-repair behaviour	P returns to about 4.5 bar and V settles to about 1.3 mm/s; HB300 back to about 0.03 mm/s [P]
Provenance	[P] runbook and equations; severity and symptoms [E]

4.2 F02 BEARING_WEAR: main bearing outer-race defect

Table 6. F02 profile

Attribute	Description
Fault	BEARING_WEAR (main bearing outer-race defect)
Affected component	Main bearing housing; main drive motor loading is secondary
Physical mechanism	A defect in the outer race produces repetitive impacts at non-integer multiples of the shaft frequency; narrow lines carry the energy, so overall RMS stays modest [P]
Sensor behaviour	V rises slowly to about 2.9 mm/s mean (peaks about 3.2 mm/s); current up about 8 percent; pressure unchanged [P]
Expected telemetry pattern	Discrete lines at about 88, 175, 263, 351 and 439 Hz (bins 7, 14, 21, 28, 35) with sidebands at plus and minus 1X; PMR300 above 3; P normal [P] / [D]
Thresholds crossed	V samples flicker over 2.5 mm/s from r about 0.6 and are over it most of the time from r about 0.8; the 4.0 mm/s alarm is never reached; the current warn limit is never reached (max 164.1 A) [P] / [D]
Severity	High: planned change costs hours, a bearing that fails in service costs days [P]
Likely causes	Fatigue, lubricant starvation or contamination history, misalignment, overload, installation damage [E]
Symptoms	Rising noise at the drive end, slight warming [E]
Alternative explanations	Imbalance or misalignment (F10, integer orders), looseness (F11), process load (F09), vibration sensor fault (F13) [E]
Differentiating evidence	Pressure normal; discrete family at non-integer orders (3.55, 7.1, 10.65, 14.2, 17.75); defect bins each more than 4 times healthy; I60 up about 4.8 A per mm/s of V rise [P] / [D]
Inspection requirements	Schedule an inspection at the next planned stop and confirm the defect frequencies with a handheld analyser [P]
Corrective action	Plan a bearing replacement; use the specified bearings and fits; lock-out and tag-out before opening the main drive [P]
Expected post-repair behaviour	V settles to about 1.3 mm/s; the harmonic family disappears; current returns to about 94.8 A mean [P] / [D]
Provenance	[P] runbook, equations and tests; symptoms and causes [E]

4.3 F03 Lube pump wear or failure

Table 7. F03 profile

Attribute	Description
Fault	Lube pump wear or failure
Affected component	Gear pump in the lubrication unit
Physical mechanism	Internal wear lowers delivery; worn or failed gears reduce pressure and change the 1 Hz pulsation
Sensor behaviour	P falls, pulsation amplitude and shape change; V rises later through starvation; I may fall slightly with the pump load
Expected telemetry pattern	P60 falls with irregular or reduced ripple; no HB300 rise early; V60 rises only after the pressure has fallen
Thresholds crossed	P crosses 3.6 and 3.0 bar; V follows [I]
Severity	High (loss of lubrication)
Likely causes	Wear, cavitation, coupling failure, low oil level
Symptoms	Low pressure, pump noise [E]
Alternative explanations	Clogged filter (F01), leak (F04), relief valve (F06)
Differentiating evidence	Pressure fall without an early broadband floor; ripple amplitude below 0.12 bar or irregular; filter swap does not restore pressure
Inspection requirements	Pump drive and coupling, delivery test, suction line, filter delta-P with a gauge (not specified in repository)
Corrective action	Repair or replace the pump per OEM; check suction
Expected post-repair behaviour	Ripple 0.12 bar at 1 Hz and P about 4.5 bar return
Provenance	[E]; numbers [I]

4.4 F04 Oil leak or low reservoir level

Table 8. F04 profile

Attribute	Description
Fault	Oil leak or low reservoir level
Affected component	Reservoir, lines, fittings
Physical mechanism	Oil lost from the circuit; the pump draws air; pressure becomes low and unsteady
Sensor behaviour	P falls and fluctuates with noise above 0.03 bar; V rises through starvation; spectrum may show broadband energy
Expected telemetry pattern	P60 falls irregularly with intermittent drops; noise std dev above healthy; later V60 rise
Thresholds crossed	P crosses 3.6 and 3.0 bar intermittently [I]
Severity	High
Likely causes	Leaking seal or fitting, missed top-up, damaged line
Symptoms	Oil on the floor, low level [E]
Alternative explanations	Clogged filter (F01), pump wear (F03)
Differentiating evidence	Irregular pressure with aeration noise and a level drop; a new filter does not restore pressure
Inspection requirements	Reservoir level, leak search along lines and fittings
Corrective action	Repair leak, refill with specified oil, bleed air
Expected post-repair behaviour	Stable P about 4.5 bar with 0.03 bar noise
Provenance	[E]; numbers [I]

4.5 F05 Blocked nozzle or line after the transducer

Table 9. F05 profile

Attribute	Description
Fault	Blocked nozzle or line after the transducer
Affected component	Distribution line or nozzle downstream of LUBE_OIL_PRESSURE sensor
Physical mechanism	A restriction after the transducer raises the pressure the transducer sees while less oil reaches the bearing
Sensor behaviour	P rises above the healthy band; V may rise later; spectrum like starvation if flow is lost
Expected telemetry pattern	P60 above 4.65 bar and rising, V60 rising slowly; the opposite direction to F01
Thresholds crossed	No project high limit; proposed watch above 5.0 bar [I]
Severity	Medium to high
Likely causes	Contaminated oil, collapsed hose, blocked nozzle
Symptoms	Bearing noise despite apparently good pressure [E]
Alternative explanations	Relief valve stuck closed (F06), cold oil (F07), transducer offset (F12)
Differentiating evidence	P rises while V rises; the repository calls this a different fault from the clog [P]
Inspection requirements	Nozzles and lines after the transducer
Corrective action	Clear or replace the restricted element
Expected post-repair behaviour	P returns to about 4.5 bar and V to about 1.3 mm/s
Provenance	[P] named; signature [E]

4.6 F06 Relief valve stuck or setpoint drift

Table 10. F06 profile

Attribute	Description
Fault	Relief valve stuck or setpoint drift
Affected component	Pressure relief valve in the lubrication unit
Physical mechanism	A valve stuck open dumps oil and holds pressure low; stuck closed raises pressure
Sensor behaviour	P steps to a new steady level; V and spectrum unchanged at first
Expected telemetry pattern	Step change in P with no gradual ramp and no change in spectrum shape
Thresholds crossed	P below 3.6 bar (open) or above 5.0 bar (closed) [I]
Severity	Medium
Likely causes	Contamination, spring fatigue, wrong setting
Symptoms	Pressure reading off the usual value [E]
Alternative explanations	Pump wear (F03), clogged filter (F01)
Differentiating evidence	Step-like change rather than a ramp; no broadband floor; filter swap ineffective
Inspection requirements	Valve and its setting
Corrective action	Clean, reset or replace the valve per OEM
Expected post-repair behaviour	P returns to about 4.5 bar
Provenance	[E]; numbers [I]

4.7 F07 Oil temperature or viscosity effect

Table 11. F07 profile

Attribute	Description
Fault	Oil temperature or viscosity effect
Affected component	Oil in the circuit
Physical mechanism	Viscosity changes with temperature, shifting the pressure at the same flow
Sensor behaviour	P follows a slow temperature cycle; V and spectrum unchanged
Expected telemetry pattern	Slow periodic P variation with the ambient or warm-up cycle; no V rise
Thresholds crossed	Possible low-pressure warn on a cold start [I]
Severity	Low
Likely causes	Cold start, hot ambient, cooler fault
Symptoms	Pressure differing between morning and afternoon [E]
Alternative explanations	Clogged filter (F01), pump wear (F03)
Differentiating evidence	Reversible with temperature; V unchanged; HB300 unchanged
Inspection requirements	Oil temperature (not measured in the repository)
Corrective action	No repair if within normal swing; fix cooler if hot
Expected post-repair behaviour	P returns when temperature normalises
Provenance	[E]; numbers [I]

4.8 F08 Main drive motor or supply fault

Table 12. F08 profile

Attribute	Description
Fault	Main drive motor or supply fault
Affected component	Main drive motor and its supply
Physical mechanism	Winding, supply imbalance or drive faults change the current waveform and add electromagnetic vibration
Sensor behaviour	I rises or the idle level shifts; ripple changes; spectrum shows lines at twice the supply frequency and 1X sidebands; P normal
Expected telemetry pattern	I60 up by more than 8 percent with V60 rising; lines near 100 Hz (50 Hz supply) [E]
Thresholds crossed	I crosses 175 A at advanced stage [I]
Severity	High
Likely causes	Winding degradation, phase imbalance, drive tuning
Symptoms	Motor heating, trips [E]
Alternative explanations	Process load (F09), bearing wear (F02)
Differentiating evidence	Current rises disproportionately to V (more than 4.8 A per mm/s) and line-frequency lines appear
Inspection requirements	Insulation test, supply balance, drive parameters
Corrective action	Repair or replace motor or correct the supply
Expected post-repair behaviour	I back to 94.8 A mean
Provenance	[E]; numbers [I]

4.9 F09 Process or die overload

Table 13. F09 profile

Attribute	Description
Fault	Process or die overload
Affected component	Die, press stroke, motor load
Physical mechanism	Thicker blank, die wear or setup raises the load pulse
Sensor behaviour	l _{pk} rises, working-stroke vibration swells (62.5 to 150 Hz); P and mean V little changed
Expected telemetry pattern	l _{pk} above 160 A, I60 up, V swell in the working stroke only; idle current about 88 A
Thresholds crossed	l _{pk} crosses 175 or 195 A [P]
Severity	Medium
Likely causes	Die change, material variation, lubrication of the process
Symptoms	Heavier stroke sound [E]
Alternative explanations	Motor fault (F08)
Differentiating evidence	Idle level unchanged and increase confined to the working stroke
Inspection requirements	Die setup, material, shut height
Corrective action	Correct setup or material
Expected post-repair behaviour	l _{pk} about 150 A
Provenance	[E]; limits [P]

4.10 F10 Imbalance or misalignment

Table 14. F10 profile

Attribute	Description
Fault	Imbalance or misalignment
Affected component	Shaft, coupling, flywheel
Physical mechanism	Mass imbalance or shaft misalignment excites 1X, 2X and integer orders
Sensor behaviour	V rises; A1X and A2X rise; P normal
Expected telemetry pattern	Bin 2 and bin 4 grow, no non-integer family, no broadband floor
Thresholds crossed	V60 may reach 2.5 mm/s [I]
Severity	Medium
Likely causes	Wear, loosened coupling, installation
Symptoms	Vibration at running speed [E]
Alternative explanations	Bearing wear (F02), looseness (F11)
Differentiating evidence	Energy at integer orders, not at 3.55, 7.1 ... x 1X
Inspection requirements	Coupling, alignment, balance
Corrective action	Re-align or balance
Expected post-repair behaviour	1X amplitude about 1.05 mm/s
Provenance	[E]; numbers [I]

4.11 F11 Mechanical looseness

Table 15. F11 profile

Attribute	Description
Fault	Mechanical looseness
Affected component	Bearing housing, foundation bolts
Physical mechanism	Looseness generates many harmonics of 1X and sub-harmonics, plus impacts
Sensor behaviour	V rises; spectrum shows 1X to 10X harmonics and a raised floor at low frequency
Expected telemetry pattern	Integer harmonic series; sub-harmonics at 0.5X; variable amplitude
Thresholds crossed	V60 may reach 2.5 or 4.0 mm/s [I]
Severity	Medium to high
Likely causes	Loose bolts, cracked foundation
Symptoms	Rattle [E]
Alternative explanations	Bearing wear (F02), imbalance (F10)
Differentiating evidence	Integer orders and 0.5X lines instead of non-integer family
Inspection requirements	Fasteners, foundation
Corrective action	Tighten, repair
Expected post-repair behaviour	Harmonic series disappears
Provenance	[E]; numbers [I]

4.12 F12 Pressure transducer fault

Table 16. F12 profile

Attribute	Description
Fault	Pressure transducer fault
Affected component	LUBE_OIL_PRESSURE sensor, wiring
Physical mechanism	Sensor drift, offset, blocked port or wiring fault gives a wrong reading
Sensor behaviour	P flat, offset or drifting; ripple and noise missing or changed; V, I and spectrum unchanged
Expected telemetry pattern	P changes while V60 and HB300 do not; ripple 0.12 bar missing; std dev near zero
Thresholds crossed	P below 3.0 bar with V normal [I]
Severity	Low (data)
Likely causes	Ageing, port blocked, loose connector
Symptoms	Pressure alarm without vibration [E]
Alternative explanations	Pump failure (F03), leak (F04), clog (F01)
Differentiating evidence	No corroboration from V and HB300; ripple and noise signature wrong
Inspection requirements	Transducer, port, wiring against a reference gauge
Corrective action	Replace or recalibrate the transducer
Expected post-repair behaviour	P with ripple 0.12 bar and noise 0.03 bar
Provenance	[E]; numbers [I]

4.13 F13 Vibration sensor or mounting fault

Table 17. F13 profile

Attribute	Description
Fault	Vibration sensor or mounting fault
Affected component	BEARING_VIBRATION_RMS sensor, mounting, cable
Physical mechanism	A loose mount, damaged cable or interference gives spikes, offsets or a flat signal
Sensor behaviour	V jumps or flat; RMS and spectrum disagree; spectrum shows spikes or mains lines
Expected telemetry pattern	V10 at or above 7 mm/s or zero; no change in P or I; RMS differs from the root-sum-square of the bins
Thresholds crossed	V above 4.0 mm/s with P normal and flat spectrum [I]
Severity	Low (data)
Likely causes	Loose mount, cable fault, EMI
Symptoms	Spurious vibration alarm [E]
Alternative explanations	Bearing wear (F02)
Differentiating evidence	RMS and spectrum disagree; changes not seen in I
Inspection requirements	Mount, cable, shielding
Corrective action	Refit, replace cable
Expected post-repair behaviour	V about 1.3 mm/s with normal breathing
Provenance	[E]; numbers [I]

4.14 F14 Current sensor fault

Table 18. F14 profile

Attribute	Description
Fault	Current sensor fault
Affected component	MAIN_MOTOR_CURRENT sensor
Physical mechanism	Offset, flat output or missing working-stroke pulse
Sensor behaviour	I flat or offset; pulse of +62 A missing; ripple missing
Expected telemetry pattern	I near constant; V and P unchanged
Thresholds crossed	I below 60 A while cycling [I]
Severity	Low (data)
Likely causes	Loose burden, failed transducer
Symptoms	Current reads zero while the press runs [E]
Alternative explanations	Motor stopped
Differentiating evidence	Press is cycling (V breathing at 4.6 s) while I is flat
Inspection requirements	Sensor, wiring
Corrective action	Replace or recalibrate
Expected post-repair behaviour	Pulse to about 150 A returns
Provenance	[E]; numbers [I]

4.15 F15 Data-path or gateway fault

Table 19. F15 profile

Attribute	Description
Fault	Data-path or gateway fault
Affected component	ESP32 gateway, Wi-Fi, MQTT broker, backend
Physical mechanism	Link, credential, clock or broker problems cause stale data, gaps or dropped messages
Sensor behaviour	stale, offline, seq gaps, validation drops; flat repeated values
Expected telemetry pattern	No telemetry for max(5 s, 4 x intervalMs); gaps in seq; new bootId with seq at 0
Thresholds crossed	Not applicable
Severity	Low (data)
Likely causes	Wi-Fi, credentials, TLS clock, NTP blocked
Symptoms	Twin shows nothing or stale [P]
Alternative explanations	Any mechanical fault masked
Differentiating evidence	Gateway diag fields and status
Inspection requirements	See the data-path table (Document 04)
Corrective action	Fix link; reboot gateway
Expected post-repair behaviour	status online, no gaps
Provenance	[P] data-path table

4.16 F16 Combined clog and bearing wear (not modelled)

Table 20. F16 profile

Attribute	Description
Fault	Combined clog and bearing wear
Affected component	Lube filter and main bearing
Physical mechanism	A clogged filter starves the bearing, which then wears [P]
Sensor behaviour	P falls (clog); V rises more than either fault alone; discrete lines superimposed on a broadband floor; I up between 3 and 8 percent
Expected telemetry pattern	P60 falls and HB300 rises while defect bins appear above the floor and PMR300 sits between 2.5 and 3
Thresholds crossed	Both warn limits crossed; alarm possible [I]
Severity	High
Likely causes	Neglected clog
Symptoms	Mixed vibration pattern [E]
Alternative explanations	Single faults
Differentiating evidence	Pressure fall plus lines; delta I over delta V between 0.77 and 4.8 A per mm/s
Inspection requirements	Filter first, then bearing inspection
Corrective action	Replace filter, then inspect and plan bearing replacement
Expected post-repair behaviour	P about 4.5 bar and V about 1.3 mm/s after both
Provenance	[E]; not modelled in the repository

5. Anatomy of CLOGGED_FILTER (F01)

5.1 Progression with severity

Table 21. CLOGGED_FILTER progression over 600 s per case [D]

r	P min / mean / max (bar)	V min / mean / max (mm/s)	V samples at or over warn	V samples in alarm	HB300 (mm/s)
0.00	4.40 / 4.50 / 4.59	1.18 / 1.31 / 1.72	0%	0%	0.030
0.25	3.89 / 3.99 / 4.08	2.09 / 2.24 / 2.57	1%	0%	0.283
0.30	3.78 / 3.89 / 3.99	2.26 / 2.42 / 2.76	12%	0%	0.318
0.44	3.48 / 3.61 / 3.72	2.72 / 2.93 / 3.29	100%	0%	0.409
0.50	3.36 / 3.49 / 3.60	2.92 / 3.15 / 3.52	100%	0%	0.447
0.74	2.84 / 3.00 / 3.15	3.72 / 4.03 / 4.44	100%	62%	0.594
1.00	2.29 / 2.47 / 2.67	4.58 / 4.98 / 5.45	100%	100%	0.749

5.2 Order of events

Table 22. Order of events as r rises [P] / [D]

r (approx.)	Event	Channel	Level reached
0.05	V60 reaches the Watch band 1.50 mm/s	V	L1 [I]
0.2	HB300 exceeds 8 x healthy (0.24 mm/s)	Spectrum	L1 [P] test
0.25 to 0.30	First V samples at or over 2.5 mm/s (1 percent then 12 percent)	V	Platform warn flicker
0.33	V60 reaches 2.5 mm/s	V	L2
0.44	P60 reaches 3.6 bar; all V samples at or over warn	P, V	L2 on both
0.74	P60 reaches 3.0 bar; V60 reaches 4.0 mm/s (62 percent of samples in alarm)	P, V	L3
0.99	P60 reaches 2.5 bar	P	L4 [I]
1.00	Final state: P 2.47 bar, V 4.98 mm/s, all samples in alarm	P, V	L3 to L4

Vibration warns before pressure does (about $r = 0.30$ against 0.44) [P]. A diagnosis waiting for the pressure warning loses about 14 percent of the ramp; at the default 120 s ramp that is about 17 s.

5.3 What the spectrum adds

Table 23. Spectrum of the clogged filter [P]

Property	Value
Location	Every bin from 300 Hz (bin 24) to the top of the span
Shape	Flat floor with 18 percent per-bin jitter; no discrete peaks
Level	Overall RMS scales by $(1 + 2.75 r)$; HB300 from 0.030 to 0.749 mm/s
Tests	High-band mean more than 8 x healthy; peak-to-median above 300 Hz below 2.5
Contrast with bearing wear	Bearing wear has peak-to-median above 3 and lines at fixed orders

5.4 Lead times of detection rules

Table 24. Detection lead of the proposed rules over the platform warning (sample-level) for CLOGGED_FILTER [D] / [I]

Rule	Triggers at r	Lead over platform warn ($r = 0.25$)	Lead at 120 s ramp	Lead at 600 s ramp
V60 at or above 1.50 mm/s (Watch)	0.05	0.20	24 s	120 s
HB300 at or above 0.24 mm/s	0.20	0.05	6 s	30 s
P60 at or below 4.30 bar (Watch)	0.10	0.15	18 s	90 s
V60 at or above 2.5 mm/s	0.33	-0.08	-10 s	-48 s
P60 at or below 3.6 bar	0.44	-0.19	-23 s	-114 s

6. Anatomy of BEARING_WEAR (F02)

6.1 Progression with severity

Table 25. BEARING_WEAR progression over 600 s per case [D]

r	V min / mean / max (mm/s)	V samples at or over warn	I mean / max (A)	P mean (bar)
0.00	1.18 / 1.31 / 1.72	0%	94.8 / 152.4	4.50
0.25	1.57 / 1.71 / 2.08	0%	96.6 / 154.9	4.50
0.50	1.96 / 2.11 / 2.44	0%	98.5 / 158.0	4.50
0.60	2.11 / 2.27 / 2.59	2%	99.3 / 159.2	4.50
0.80	2.40 / 2.58 / 2.89	88%	100.8 / 161.6	4.50
1.00	2.68 / 2.89 / 3.21	100%	102.3 / 164.1	4.50

6.2 Why a threshold alarm misses it

Table 26. Threshold behaviour of BEARING_WEAR [P]

Limit	Reached?	Detail
V warn 2.5 mm/s	Yes, late	Flickers from r about 0.6; over it most of the time from r about 0.8
V alarm 4.0 mm/s	Never	Peak about 3.2 mm/s at full severity (asserted by test)
I warn 175 A	Never	Maximum 164.1 A
P warn 3.6 bar	Never	P stays normal (mean 4.5 bar)

A bearing can be badly worn while RMS stays under the alarm limit because the energy sits in a few narrow lines. Only trend (drift) and spectrum shape catch it [P].

6.3 Lead times of detection rules

Table 27. Detection lead of the proposed rules over the first platform warning (r = 0.6) for BEARING_WEAR [D] / [I]

Rule	Triggers at r	Lead (r)	Lead at 120 s ramp	Lead at 600 s ramp
V60 at or above 1.50 mm/s (Watch)	0.12	0.48	58 s	288 s
Defect lines at least 4 x healthy and PMR300 above 3	Not stated in repository	Not specified	Not specified	Not specified
I60 at or above 99.5 A (1.05 x)	0.62	-0.02	-2 s	-12 s
Alarm (4.0 mm/s)	never	none	none	none

7. Discrimination of the Two Simulated Faults

7.1 Discrimination table

Table 28. Evidence that separates CLOGGED_FILTER from BEARING_WEAR (both runbooks) [P]

Evidence	Clogged lube filter	Bearing wear
Lube oil pressure	Falls	Normal
Spectrum above 300 Hz	Broadband floor, no discrete peaks	Harmonic family at fixed multiples of 1X, with sidebands
Motor current	Up about 3 percent	Up about 8 percent
Vibration RMS at full severity	Above alarm (about 5 mm/s)	Below alarm (peaks about 3.2 mm/s)
Cheapest next step	Replace the filter cartridge	Plan a bearing replacement

7.2 Quantitative thresholds for differentiation

Table 29. Differentiating bands [D] / [I]

Quantity	Points to F01	Points to F02	Ambiguous
P60	At or below 4.30 bar	4.45 to 4.55 bar	4.30 to 4.45 bar
PMR300	Below 2.5	Above 3	2.5 to 3
HB300	At or above 0.24 mm/s	Near 0.03 to 0.10 mm/s	0.10 to 0.24 mm/s
Delta I60 / Delta V60	About 0.8 A per mm/s	About 4.8 A per mm/s	1.5 to 3.5 A per mm/s
V60 above 4.0 mm/s	Yes	Never	Not applicable
Defect bins at least 4 x healthy	No	Yes (all five)	Some only

7.3 Discriminator matrix across the fault library

Codes: U up, D down, S same as healthy, F flat or implausible, P present, A absent, ? depends. Faults F03 to F16 are [E].

Table 30. Discriminator matrix

Evidence	F01	F02	F03	F04	F05	F08	F09	F10	F12	F13
P60	D	S	D	D	U	S	S	S	D or F	S
V60	U	U	U late	U late	U late	U	U	U	S	U or F
I60	U small	U	D or S	S	S	U	U	S	S	S
Ipk (stroke peak)	S	S	S	S	S	U	U	S	S	S
HB300 broadband	U	S	U late	U late	U late	S	S	S	S	P spikes
Discrete non-integer lines	A	P	A	A	A	A	A	A	A	A
Integer orders 1X to 10X	A	A	A	A	A	A	A	P	A	?
Pressure ripple 0.12 bar	P	P	Change d	Irregular	P	P	P	P	Missing	P
Line-frequency lines	A	A	A	A	A	P	A	A	A	?
RMS equals spectrum	P	P	P	P	P	P	P	P	P	A

8. Pattern-to-Fault Mapping

Table 31. Sensor pattern, possible fault, evidence, severity, inspection and action

Sensor pattern	Possible fault	Evidence	Severity	Inspection	Action
P60 falling, V60 rising, flat high-band floor	F01 CLOGGED_FILTER	HB300 up, PMR300 below 2.5, I60 up 3 percent	Medium to high	Filter cartridge, line pressure	Replace cartridge
V60 rising slowly, P normal, discrete lines	F02 BEARING_WEAR	Lines at 88, 175, 263, 351, 439 Hz; I60 up about 8 percent	High	Handheld analyser at the housing	Plan bearing change
V60 at 2.5 to 3.5 mm/s, P normal, no lines	F10 or F11 or F13	Integer orders or RMS mismatch	Medium	Alignment, fasteners, sensor mount	Correct
P low, V normal	F12, F03, F04 or F06	No broadband floor, no V rise	Medium	Transducer then pump and leak	Verify sensor first
P high	F05 or F06	P above 5.0 bar [I]	Medium	Nozzles and lines, relief valve	Clear restriction
Ipk at or above 175 A, idle level normal	F09	Increase only in working stroke	Medium	Die, material	Correct setup
I60 up more than 8 percent, line-frequency lines	F08	I per mm/s above 4.8 A	High	Motor and supply	Repair motor
Pressure low and discrete lines both	F16	Mixed evidence	High	Lubrication first, then bearing	Filter, then plan bearing
Flat or implausible channel	F12, F13, F14	No cross-channel confirmation	Low (data)	Sensor and wiring	Repair sensor
Stale, gaps, offline	F15	diag, status	Low (data)	Gateway and network	Fix link

9. Symptom-to-Cause Mapping

Table 32. Observed symptom to candidate causes

Symptom	First candidate	Second candidate	Rule out first
Vibration warning	F01 if pressure is falling	F02 if pressure is normal	F13 sensor, F15 data
Vibration warning with pressure warning	F01	F16	F12 transducer
Pressure warning, vibration normal	F12 transducer	F03, F04, F06	F15 data
Current warning	F09 process	F08 motor	F14 sensor
Slow V rise over hours, below warn	F02	F01 early	Baseline error
Sudden step in V	F13 mount	F11 looseness	Process change
Pressure ripple missing	F12	Sampling alias	Data path
Alarm clears and returns	Hysteresis effects	BEARING_WEAR flicker (r 0.6 to 0.8)	Per-sample classification

10. Severity Classification

Table 33. Severity classes

Class	Definition	Examples	Response
S1 Low	No risk to the machine; data or minor condition	F07, F12 to F15	Fix at convenience; verify data
S2 Medium	Degradation, cheap to fix if early	F01 at r below 0.74, F05, F06, F09, F10	Next planned stop
S3 High	Risk of damage and costly stop if ignored	F02, F03, F04, F08, F16, F01 at alarm	Plan repair this shift
S4 Critical	Immediate damage risk	Any fault at L4	Controlled stop

Table 34. Severity versus level and growth

Fault	L1	L2	L3	L4
F01 CLOGGED_FILTER	S2	S2	S3	S4 if P60 at or below 2.5 bar
F02 BEARING_WEAR	S2	S3	S3	S4 at V10 at or above 7 mm/s
F03 to F04	S3	S3	S4	S4
F12 to F15	S1	S1	S1	S1

11. Alternative Explanations and Inspection Requirements

Table 35. Alternatives that must be excluded before confirming a fault

Leading hypothesis	Alternative	How to exclude	Inspection that confirms
F01	F03 pump wear	Pulsation 0.12 bar present and a flat broadband floor; fall consistent with delta V / delta P of -1.8	Filter cartridge appearance; pressure after replacement
F01	F12 transducer	V60 and HB300 confirm the fall	Reference gauge reading
F02	F10 or F11	Lines at non-integer orders only	Handheld analyser order analysis
F02	F13 sensor	RMS equals spectrum; current moves with V	Re-mount sensor and compare
F02	F01 early	P60 at or above 4.45 bar; HB300 below 0.10 mm/s	Pressure trend over the last hour
F09	F08	Idle level unchanged	Current at idle and in stroke
F05	F12	V rising too	Reference gauge, nozzle check
F16	F01 plus noise	Lines above floor, PMR300 between 2.5 and 3	Filter then re-check spectrum

Table 36. Inspection requirements by fault

Fault	Inspection	Tool	Pass criterion [I]
F01	Filter cartridge, line pressure	OEM service tools	Clean cartridge; pressure returns to 4.5 bar
F02	Bearing frequencies, housing condition	Handheld analyser	No lines at the defect orders; V 1.3 mm/s
F03 to F04	Pump, reservoir level, leaks	OEM tools	Level correct, no leaks
F08	Insulation and supply balance	Electrical test set	Within OEM limits
F12 to F14	Sensor against a reference	Reference gauge, clamp meter, analyser	Agrees within tolerance

12. Summary, Safety, References and Revision

12.1 Diagnostic summary

- Two simulated faults, one shared symptom (high bearing vibration): pressure, spectrum shape, current ratio and alarm behaviour tell them apart [P].
- CLOGGED_FILTER: P60 falls to about 2.47 bar, V60 reaches about 5 mm/s, flat high-band floor, I up 3 percent.
- BEARING_WEAR: P normal, V60 reaches about 2.9 mm/s (peaks 3.2), discrete family at 88, 175, 263, 351, 439 Hz, I up 8 percent, alarm never reached.
- Mixed evidence means lubrication first [P]. F03 to F16 are engineering reference [E]; numbers [I].

12.2 Key thresholds

Table 37. Key thresholds for fault recognition

Quantity	Value	Tag
V warn / alarm	2.5 / 4.0 mm/s	[P]
P warn / alarm	3.6 / 3.0 bar	[P]
I warn / alarm	175 / 195 A	[P]
HB300 test	above 0.24 mm/s (8 x healthy)	[P] test
PMR300	below 2.5 flat; above 3 peaked	[P] tests
Defect bins	7, 14, 21, 28, 35 at least 4 x healthy	[P] test
Critical (V, P, I)	7.0 mm/s, 2.5 bar, 220 A	[I]

12.3 Safety and escalation notes

This document is a capstone prototype, not an OEM manual. Apply lock-out and tag-out and follow the manufacturer's procedure before opening the lubrication circuit or the main drive. The agent cannot act on the machine.

12.4 References and engineering basis

- CONTEXT_PRESS_STAMP_01.md, firmware/catalog.json, simulation.h, sim-models.mjs, docs/mqtt-contract.md, the two press runbooks, simulator.test.js [P].
- General vibration-diagnostics practice for rolling-element bearings (non-integer defect orders, sidebands at shaft speed) and lubrication-circuit practice [E].

12.5 Revision notes

Table 38. Revision history

Rev	Date	Change
1.0	2026-10-08	First issue